

Repeat without rewriting



Loop idea

```
i = 1
while i <= 5:
    print("Hello World")
    i += 1
```

while runs a block while a condition is true.

Perfect when you need controlled repetition.

The problem with copy-paste

Repeated code

```
print("Hello World")  
print("Hello World")  
print("Hello World")  
print("Hello World")  
print("Hello World")
```

With while

```
i = 1  
while i <= 5:  
    print("Hello World")  
    i += 1
```

Loops replace repeated code with one clear pattern.

3 parts

make it work



while loop anatomy

```
i = 1                # initialization
while i <= 5:        # condition
    print("Hello World")
    i += 1           # increment
```

Initialize

Create the counter before the loop starts.

Check

Python tests the condition before each run.

Update

Change the counter so the loop can stop.

Debugger

view

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Step by step

```
i = 1
while i <= 5:
    print("Hello")
    i += 1
# i becomes 6
# condition false
```

Counter trace

Run	i	i <= 5
1	1	True
2	2	True
3	3	True
4	4	True
5	5	True
Exit	6	False

When the condition becomes false, Python exits the loop.

Nested while loops



Outer loop

```
i = 1
while i <= 3:
    print("Telusko")
    j = 1
    while j <= 2:
        print("rocks")
        j += 1
    i += 1
```



Output rhythm

```
Telusko
rocks
rocks
Telusko
rocks
rocks
Telusko
rocks
rocks
```

The inner loop completes every time the outer loop runs.

Reset the inner counter



Counter placement

```
day = 1
while day <= 3:
    hour = 1      # reset for each day
    while hour <= 24:
        print(day, hour)
        hour += 1
    day += 1
```

Outer

Think of days: day 1, day 2, day 3.

Inner

Think of hours: 1 to 24 inside each day.

Reset

Put hour = 1 inside the day loop.

Cheat Sheet

Python while loops

Basic syntax

```
counter = start
while condition:
    repeated_work()
    counter += step
```

Use for

Repeating work while a condition stays true.

Must update

Change the counter or the loop may never stop.

Debug it

Step through to watch values change.

Nested loops

Reset the inner counter inside the outer loop.

Save this for your Python journey.

Day 25/55 - Prepared by Satyanarayan Sen